

Evaluation Framework and Results for LLM Dental Patient Simulation

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1 Evaluation Methodology

To ensure the clinical accuracy and immersive quality of the virtual patient, we implemented a rigorous testing suite comprising three distinct metrics. The model was subjected to a standardized interview protocol consisting of opening questions, chief complaint inquiries, and disease-specific diagnostic questions.

1.1 Symptom Fidelity Score (SFS)

The Symptom Fidelity Score measures the accuracy of the LLM’s clinical presentation against the ground truth symptoms defined in the knowledge base. It penalizes both the omission of required symptoms and the hallucination of non-existent symptoms.

Let $S_{expected}$ be the set of symptoms required for a specific diagnosis. Let $S_{mentioned}$ be the set of symptoms extracted from the LLM’s responses, and $S_{hallucinated}$ be the subset of mentioned symptoms that are medically contraindicated or irrelevant to the condition.

The score is calculated as:

$$SFS = \max \left(0, 1 - \frac{N_{missing} + N_{hallucinated}}{|S_{expected}|} \right) \quad (1)$$

Where:

- $N_{missing} = |S_{expected} \setminus S_{mentioned}|$
- $N_{hallucinated}$ is the count of medically inaccurate additions (e.g., claiming fever when none exists).

1.2 Role-Playing Consistency Score (RPCS)

This metric evaluates the stability of the patient persona. It detects breaks in character, such as the use of medical jargon ("diagnosis", "prognosis"), providing medical advice (which a patient should not do), or "meta-awareness" (referring to itself as an AI).

Let N_{turns} be the total number of assistant responses in a conversation. Let V be the total count of violation patterns detected across all responses. The violation rate R_v is defined as V/N_{turns} .

The RPCS is a step-function defined as:

$$RPCS = \begin{cases} 1.0 & \text{if } V = 0 \\ 0.8 & \text{if } 0 < R_v \leq 0.1 \\ 0.5 & \text{if } 0.1 < R_v \leq 0.3 \\ 0.2 & \text{if } R_v > 0.3 \end{cases} \quad (2)$$

1.3 Clinical Realism Score (CRS)

The Clinical Realism Score assesses the linguistic quality of the simulation. It evaluates four dimensions, each scored on an integer scale from 0 to 4 ($S_{dim} \in [0, 4]$):

1. **Symptom Description:** Preference for subjective, concrete language (e.g., "it hurts when I bite").

2. **Vocabulary:** Usage of layperson terms vs. medical jargon.
3. **Pain Description:** Specificity regarding location, intensity, and triggers.
4. **Consistency:** Internal logical consistency of the narrative.

The normalized aggregate score is calculated as:

$$CRS = \frac{1}{16} \sum_{i=1}^4 S_{dim_i} \quad (3)$$

2 Experimental Results

The model was evaluated across 8 distinct dental pathologies. For each disease, 3 full consultation simulations were run (Total $N = 24$ tests).

2.1 Aggregate Performance

The system demonstrated high performance across all metrics, with particular strength in Role-Playing Consistency and Clinical Realism.

Table 1: Aggregate Metrics Across All Tests

Metric	Mean	Min	Max
Symptom Fidelity (SFS)	0.874	0.500	1.000
Role Consistency (RPCS)	0.958	0.800	1.000
Clinical Realism (CRS)	0.971	0.813	1.000

2.2 Disease-Specific Analysis

The following table details the average performance for each dental condition tested. High fidelity was observed in pulpal diseases (Necrosis, Reversible Pulpitis), while Periodontal Abscess showed slightly lower symptom fidelity due to the complexity of distinguishing gum vs. tooth pain descriptions.

Table 2: Performance Breakdown by Disease (Average of 3 runs)

Disease	Avg SFS	Avg RPCS	Avg CRS	Overall Score
Reversible Pulpitis	1.000	1.000	0.938	0.979
Pulp Necrosis	1.000	1.000	0.958	0.986
Irreversible Pulpitis	0.905	1.000	1.000	0.968
Simple Cavities	0.944	0.933	0.979	0.952
Pericoronitis	0.917	0.933	1.000	0.950
Chronic Apical Periodontitis	0.722	1.000	1.000	0.907
Acute Apical Periodontitis	0.833	0.933	0.896	0.888
Periodontal Abscess	0.667	0.867	1.000	0.844

2.3 Error Analysis

- **Hallucinations:** Minor hallucinations were detected in the *Periodontal Abscess* tests (e.g., mentioning nausea/fever when not present in the profile).
- **Consistency:** The model achieved a perfect RPCS score (1.0) in 5 out of 8 diseases, indicating strong persona adherence.
- **Meta-Awareness:** Violations where the model admitted to being an AI or used phrases like "I don't have" (meta-pattern) were rare, occurring in less than 7% of responses in the affected trials.